When I say AI, what do you think of?

AI is a **field of computer science** that started in the 1950s.

It includes:

- Logic
- Search algorithms
- Planning
- Expert systems
- Probability
- Machine learning
- Robotics
- Natural language processing
- AND generative AI

Generative AI is just one branch.

How can machines reason, decide, plan, and solve problems?

Examples:

- Chess engines
- Medical diagnosis systems
- Route planning (Google Maps)
- Spam detection
- Self-driving cars
- Decision trees
- Expert systems

Generating new content (text, images, audio, code).

Examples:

- ChatGPT
- Image generation
- Code generation

It is a **subset of Machine Learning**, which is a **subset of AI**.

AI
→ Machine Learning
  → Deep Learning
    → Generative AI

ChatGPT does not “think” like logic-based AI.

Earlier AI was:

- Symbolic
- Rule-based
- Logical

Now AI is often:

- Statistical
- Data-driven

An Introduction to Logics and Rule-Based (Expert) Systems

Artificial Intelligence

What is Propositional Logic?

Propositional logic involves formulas that represent propositions (statements that are either true or false).

- **Example**: "It is raining." (This can be represented by the proposition P).
- **Syntax**:
  - Variables (e.g., P, Q).
  - Logical connectives: AND (∧), OR (∨), NOT (¬) IMPLIES (→).
- **Truth Table**: Example of a truth table for P∧Q.

A truth table is used to show the truth values of a logical expression for all possible truth values of the variables.

Satisfiability and Validity of Logical Formulas

- A logical formula is satisfiable if there is some assignment of truth values to its variables that makes the formula true.

- A logical formula is valid if it is true for all possible truth assignments.
- Example:
    - $P \rightarrow Q$ is valid when P is true, and Q is true.
    - $P \rightarrow Q$ is satisfiable when P=true and Q= false

Role of Logic in AI

- Logical reasoning is used in AI to make decisions, deduce facts, and infer consequences.
- Example: In autonomous driving, logic is used to infer that if the road is blocked (proposition), the car should stop.

- **Applications**: Expert systems, automated theorem proving, natural language understanding.

Basics of Logic and Reasoning in AI

- **Logic** is the foundation of AI systems, allowing them to make decisions based on reasoning.
- In AI, we often use **propositional logic** and **predicate logic** to represent knowledge and perform reasoning tasks.
- **Reasoning** is the process of deriving conclusions from premises. In AI, this means making decisions, predictions, and inferences based on data.
- **Example**:
    - If we have the statements: "If it is raining, then the ground is wet" (If P then Q), we can deduce that if P is true (it’s raining), then Q must also be true (the ground is wet).

Propositional Logic

- Statements that are either true or false. It uses logical connectives such as AND (∧), OR (∨), NOT (¬), and IMPLIES (→).
- Each proposition is a simple statement like "It is raining" or "The ground is wet.“

- **Logical connectives** allow combining simple statements to form more complex logical expressions.
- **Example**:
    - "It is raining" → P
    - "The ground is wet" → Q
    - The logical formula "P → Q" means "If it is raining, then the ground is wet."

Example of Propositional Logic

- **Example 1**: "If it rains, then I will take an umbrella."
  - P: "It rains."
  - Q: "I will take an umbrella."
  - Formula: **P** → **Q** ("If it rains, then I will take an umbrella.")
- **Example 2**: "I will go to the beach if it is sunny or if I am on vacation."
  - P: "It is sunny."
  - Q: "I am on vacation."
  - R: "I will go to the beach."
  - Formula: **(P ∨ Q)** → **R** ("If it is sunny or I am on vacation, then I will go to the beach.")

Propositional Logic (0th Order Logic)

What it works with:

Whole statements that are either **true or false**.

You cannot look inside them.

Example:

- P = “It is raining”
- Q = “The ground is wet”

You can combine them:

- $P \rightarrow Q$
- $P \wedge Q$
- $\neg P$

- **Exercise**:
  - Write a propositional logic formula for: "If it is not raining and I am free, I will go for a walk."

- Solution: $(\neg R \land F) \rightarrow W$, where:
  - R = "It is raining"
  - F = "I am free"
  - W = "I will go for a walk"

Truth Tables

- A truth table is a systematic way of listing all possible truth values for a logical expression.
- It helps to evaluate the truth of a complex logical expression based on the truth values of its components.
- Example: For the expression $P \rightarrow Q$, the truth table is as follows:

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Exercise:

- Build the truth table for the formula **(P ∧ Q) → R**.

P	Q	R	P ∧ Q	(P ∧ Q) → R
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	T
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

Predicate Logic (First-Order Logic)

This is more powerful
we can use:
Variables
Predicates
Quantifiers

What is a predicate?

A predicate is a statement that depends on a variable.

Example:

- Human(x)
- Mortal(x)

Now we can write:

$$\forall x(Human(x) \rightarrow Mortal(x))$$

This means:

"For all x, if x is human, then x is mortal."

This was impossible in propositional logic.

Introduction to Decision Making in AI

- Decision making in AI systems involves selecting the best action or outcome from a set of possibilities.
- AI uses different techniques such as **heuristics**, **search algorithms**, and **probability** to make decisions.
- **Example**: A chess AI evaluates different possible moves based on future outcomes.
- **Practical Exercise**:
- When making a decision, should an AI prioritize **accuracy** or **speed**? Discuss the pros and cons.

In a real-time system, speed might be prioritized, whereas for medical diagnosis, accuracy is crucial.

Heuristics

Instead of checking every restaurant in the city:
You choose the one with:

- Highest rating
- Closest distance
- Most reviews

That's a heuristic.
You didn't compute the optimal restaurant mathematically.

If traffic looks heavy ahead,
you take a side street.
You didn't simulate every possible route.
You used experience.

That's heuristic reasoning.

- **Uncertainty** arises when we don’t have complete information about the environment.

- **Uncertainty** in AI refers to situations where we can’t predict the exact outcome due to incomplete, ambiguous, or noisy data.

- In AI, we deal with uncertainty using **probability** and **statistical models**.
    - **Examples**:
        - A self-driving car has sensors to detect pedestrians, but at night or in foggy conditions, the sensors may be less accurate. The system can’t be sure if there is a pedestrian, leading to uncertainty in decision-making.

        - In a recommendation system, if a user is new to the platform, there’s uncertainty in predicting their preferences because of the lack of prior data.

- **Exercise**:
  - Give an example of a situation in AI where uncertainty must be handled (e.g., recommendation systems, weather forecasting, autonomous vehicles).

Solution: Autonomous vehicles must handle uncertainty due to noisy sensor data in complex environments like heavy rain or fog.

- A **random variable** is a variable whose value is determined by the outcome of a random event. These variables are used to model uncertain outcomes.
- A **probability distribution** describes the likelihood of each possible value for a random variable.
- **Example**:
  - In a coin toss, the random variable is the outcome of the toss, and it can either be heads or tails, each with a probability of 0.5.
- **Exercise**:
  - If we roll a fair 6-sided die, what is the probability distribution for the outcome?

Solution: The probability distribution is uniform, with each outcome having a probability of 1/6.

Conditional Probability and Bayes' Theorem

- **Conditional Probability**: The probability of an event occurring given that another event has already occurred.
- **Bayes' Theorem** helps update the probability of a hypothesis based on new evidence.
    - **Example 1**:
        - If the probability of getting sick is 0.1, and the probability of showing symptoms given that you're sick is 0.8, we can use Bayes' Theorem to calculate the probability of being sick given that symptoms are present.

The Formula

$$P(H \mid E) = \frac{P(E \mid H)\,P(H)}{P(E)}$$

Where:

- **H** = Hypothesis
- **E** = Evidence

And:

- $P(H)$ = Prior probability (what we believed before)
- $P(E|H)$ = Likelihood (how likely evidence is if hypothesis is true)
- $P(E)$ = Total probability of the evidence
- $P(H|E)$ = Posterior probability (updated belief)

Intuitive Example — Medical Test

Suppose:

- 1% of people have a disease
  $\rightarrow P(Disease) = 0.01$
- Test is 90% accurate if someone is sick
  $\rightarrow P(Positive|Disease) = 0.9$
- 10% false positive rate
  $\rightarrow P(Positive|NoDisease) = 0.1$

You test positive.

What is:

$$P(Disease|Positive)$$

Most people say "90%".

That's wrong.

$$P(Positive) = P(Positive|Disease)P(Disease) + P(Positive|NoDisease)P(NoDisease)$$

$$= (0.9)(0.01) + (0.1)(0.99)$$

$$= 0.009 + 0.099$$

$$= 0.108$$

Now apply Bayes:

$$P(Disease|Positive) = \frac{0.9 \times 0.01}{0.108}$$

$$\approx 0.083$$

So probability is only 8.3%.

- **Exercise**:
  - If the probability of rain on a given day is 0.4, and the probability of the forecast predicting rain given that it is going to rain is 0.9, what is the probability that it will rain if the forecast predicts rain?

- Solution: Use Bayes' Theorem:

$$P(\text{Rain} \mid \text{Forecast}) = \frac{P(\text{Forecast} \mid \text{Rain}) \times P(\text{Rain})}{P(\text{Forecast})}$$

If we assume the forecast is correct 70% of the time, then:

$$P(\text{Rain} \mid \text{Forecast}) = \frac{0.9 \times 0.4}{0.7} \approx 0.514$$

So, there's a 51.4% chance it will rain if the forecast predicts rain.

- P(Rain)=0.3 (There's a 30% chance of rain on any given day.)
- P(Forecast|Rain)=0.8 (The forecast predicts rain 80% of the time when it actually rains.)
- P(Forecast)=0.5 (The forecast predicts rain 50% of the time in general.)

- We want to calculate the probability of rain, given that the forecast predicts rain (P(Rain|Forecast).
- Using Bayes' Theorem:

$$P(\text{Rain} \mid \text{Forecast}) = \frac{P(\text{Forecast} \mid \text{Rain}) \cdot P(\text{Rain})}{P(\text{Forecast})}$$

- So, after considering the

$$P(\text{Rain} \mid \text{Forecast}) = \frac{0.8 \cdot 0.3}{0.5} = 0.48$$

.

- **Entropy** measures the uncertainty or disorder in a system. Higher entropy indicates greater uncertainty.
- It’s crucial in AI, particularly in decision trees and learning algorithms, where it helps determine how to split data.
- **Example**:
  - **Coin Toss**: In a fair coin toss, there are two equally outcomes:
    - heads and tails.
  - The entropy is maximum, reflecting the high uncertainty of the outcome.

Exercise:

- Calculate the entropy for a dataset with the following distribution: 3 heads, 1 tail.

- Solution:

$$H(S) = -P(\text{Head})\log_2 P(\text{Head}) - P(\text{Tail})\log_2 P(\text{Tail})$$

$$H(S) = -\left(\frac{3}{4}\log_2\frac{3}{4} + \frac{1}{4}\log_2\frac{1}{4}\right) \approx 0.81$$

The entropy of the dataset is 0.81 bits.

- A coin has two equally outcomes: heads (H) and tails (T). Since both outcomes have the same probability (50%), the uncertainty is at its maximum.
- **Entropy**: The entropy H(X) of a random variable X is calculated using the formula:

$$H(X) = -\sum_{i=1}^{n} P(x_i) \log_2 P(x_i)$$

outcome.
tcomes.

- For a fair co s: heads and tails. Both have a probability of 0.5, so:

$$H(\text{Coin Toss}) = -\left[P(\text{Heads}) \log_2 P(\text{Heads}) + P(\text{Tails}) \log_2 P(\text{Tails})\right]$$

$$H(\text{Coin Toss}) = -\left[0.5 \log_2 0.5 + 0.5 \log_2 0.5\right]$$

$$H(\text{Coin Toss}) = -\left[0.5 \times (-1) + 0.5 \times (-1)\right]$$

$$H(\text{Coin Toss}) = -\left[-0.5 - 0.5\right] = 1$$

So, the **entropy** of a fair coin toss is **1 bit**.

- **Problem**: Consider a system where the probability of event A is 0.6, and the probability of event B is 0.4. Calculate the entropy of the system.

Solution:

$$H(S) = -\left(P(A)\log_2 P(A) + P(B)\log_2 P(B)\right)$$

$$H(S) = -\left(0.6\log_2 0.6 + 0.4\log_2 0.4\right)$$

$$H(S) = -\left(0.6 \times (-0.736) + 0.4 \times (-1.322)\right)$$

$$H(S) = -\left(-0.4416 - 0.5288\right)$$

The entropy is approximately **0.97 bits**.

Information Gain tells us **how much uncertainty (entropy)** is **reduced** after splitting the dataset on a particular feature.

- **Information Gain** is a measure of how well a feature splits the dataset. It measures how much information is obtained by splitting a dataset based on a particular attribute.
- It's used in decision trees to choose the best feature to split on.
- A feature with higher information gain helps reduce uncertainty in the data.

In a decision tree used to classify animals, splitting the data based on the attribute "has fur" gives a significant gain in information.

- **Example**:
  - A decision tree for predicting whether a customer buys a product based on features like Age, Income, and Location. We calculate the information gain for each feature and choose the one with the highest gain to split the data.

- **Exercise**: Given a dataset of animals with attributes like “Has Fur” and “Can Fly,” which attribute would give the highest information gain for classifying the animals as "Mammal" or "Bird"?

Solution: Calculate the entropy for both attributes and choose the one with the higher reduction in uncertainty.

- **What is a Decision Tree?** flowchart-like structure that helps AI make decisions by splitting data based on attribute values.
- **When to Use**:
  - **Classification Problems**: Classifying an object into predefined categories (e.g., spam vs. non-spam emails).
  - **Regression Problems**: Predicting a continuous value (e.g., predicting house prices).
- **Example**: A decision tree used by a bank to decide whether to approve a loan based on factors like income, age, and credit score.

- **Exercise**: Create a simple decision tree to decide whether to play tennis based on the weather conditions (sunny, rainy, etc.).

  Weather (Sunny, Rainy, Overcast)

  Humidity (High, Normal)

  Wind (Strong, Weak)
- Given a dataset with historical observations, a logical decision tree could be:
  - **If the weather is "Overcast" → Play Tennis (Yes)**
  - **If the weather is "Sunny"**:
    - If humidity is **High** → Don’t Play
    - If humidity is **Normal** → Play
  - **If the weather is "Rainy"**:
    - If wind is **Strong** → Don’t Play
    - If wind is **Weak** → Play

- Entropy and Information Gain are the key concepts used to determine which attribute to split on when constructing a **Decision Tree**.

Formula for Information Gain:

$$\text{Information Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

Information Gain = Entropy before split – Weighted entropy after split

- S is the dataset.
- A is the attribute.
- S_v is the subset of data where the attribute A has value v.
- $|S_v|/|S|$ = Proportion of data in that subset

10 Students – Will Pass Exam? (Yes/No)

Dataset S:

- 5 Pass
- 5 Fail
- Total = 10

Compute Entropy Before Split

Since 5/5:

H(S)=1 Maximum uncertainty.

Feature A: "Studied?"

Feature A has two values:

- Yes
- No

After Split:

Group 1: Studied = Yes

- 4 Pass
- 1 Fail
- Total = 5

Entropy: H(Yes)≈0.72

Group 2: Studied = No

- 1 Pass
- 4 Fail
- Total = 5

Entropy: H(No)≈0.72

Compute Weighted Entropy After Split

Each group has 5 students out of 10.

$$H(S|A) = \frac{5}{10} \times 0.72 + \frac{5}{10} \times 0.72$$

$$= 0.72$$

Compute Information Gain

$$IG = H(S) - H(S|A)$$

$$IG = 1 - 0.72$$

$$IG = 0.28$$

- **What is Entropy?**
- Entropy measures the **impurity** or **uncertainty** in a dataset. It quantifies how mixed the class labels are.
  - If all examples belong to **one** class → **Entropy = 0** (pure dataset, no uncertainty).

Weather	Play Tennis?
Sunny	Yes
Sunny	Yes
Rainy	Yes
Overcast	Yes

  - If examples are evenly split s → **Entropy = 1** (maximum uncertainty). 50% "Yes", 50

Weather	Play Tennis?
Sunny	Yes
Sunny	No
Rainy	Yes
Rainy	No

so we cannot confidently predict "Yes" or "No."

- **Entropy Formula:**

p_i is the pr dat

$$H(S) = -\sum p_i \log_2 p_i$$

- Suppose we have the following dataset (Play Tennis decision based on Weather):
- **Compute Entropy of Entire Dataset**
- We count:
  - **Yes** = 9 times
  - **No** = 5 times

$$H(S) = -\left(\frac{9}{14}\log_2\frac{9}{14} + \frac{5}{14}\log_2\frac{5}{14}\right)$$

We split the data into three groups based on "Weather":

Sunny → 5 instances → (2 Yes, 3 No)

Overcast → 4 instances → (4 Yes, 0 No)

Rainy → 5 instances → (3 Yes, 2 No)

Entropy for "Sunny" (5 samples: 2 Yes, 3 No)

$$H(\text{Sunny}) = -\left(\frac{2}{5}\log_2\frac{2}{5} + \frac{3}{5}\log_2\frac{3}{5}\right) = 0.971$$

We calculate H(overcast) =0 as we have 4 Yes, 0 No

H(Rainy) =0.971

We compute the **weighted sum** of the entropy of each subset:

The IG for splitting on "Weather" is 0.246 tells us that splitting on "Weather" reduces uncertainty in the dataset by 0.247 bits.

Weather	Play Tennis?
Sunny	No
Sunny	No
Overcast	Yes
Rainy	Yes
Rainy	Yes
Rainy	No
Overcast	Yes
Sunny	No
Sunny	Yes
Overcast	Yes
Overcast	Yes
Rainy	No

- You may calculate the Information Gain (IG) for other attributes such as:
  - **"Temperature"** (Hot, Mild, Cool)
  - **"Humidity"** (High, Normal)
  - **"Wind"** (Weak, Strong)

Attribute	Information Gain (IG)
Weather	**0.247**
Humidity	0.151
Wind	0.048
Temperature	0.029

- As "Weather" ribute for the first split in the Decision Tree. ches...